SIMATS ENGINEERING

ASSESSMENT TOOL 1 - ANNEXURE A

Analytical Problem Solving: Questions with Answers

Course Name: Computer Networks	**Course Code:** CSA07
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Course Outcome: CO2 (BL3)	**Marks:** 8 per question

1. A university network is assigned the block 192.168.20.0/24. Lab A uses a /26 subnet, Lab B a /27 subnet and Lab C a /28 subnet. Find the network address, valid host range and broadcast address of each lab, and the number of usable hosts. (8 Marks)

Problem Understanding

Given block 192.168.20.0/24 (256 addresses). Required: network address, host range, broadcast address and usable hosts for /26, /27 and /28 subnets.

Method Selection

- Block size = $2^{(32 - \text{prefix})}$; usable hosts = $2^{h} - 2$, where h = host bits.
- Network address = first address of block; broadcast = last address; host range = network + 1 to broadcast - 1.
- Subnets are allocated one after another (largest first) so that they do not overlap.

Solution Process

- /26: mask 255.255.255.192, block = 64, host bits = 6, hosts = $2^{6} - 2 = 62$.
- /27: mask 255.255.255.224, block = 32, host bits = 5, hosts = $2^{5} - 2 = 30$.
- /28: mask 255.255.255.240, block = 16, host bits = 4, hosts = $2^{4} - 2 = 14$.

Lab	Subnet	Network address	Valid host range	Broadcast	Usable hosts
A	/26	192.168.20.0	192.168.20.1 - 192.168.20.62	192.168.20.63	62
B	/27	192.168.20.64	192.168.20.65 - 192.168.20.94	192.168.20.95	30
C	/28	192.168.20.96	192.168.20.97 - 192.168.20.110	192.168.20.111	14

Interpretation of Results

The three subnets are contiguous and non-overlapping, using 64 + 32 + 16 = 112 addresses (192.168.20.0 to .111). The range 192.168.20.112 - 192.168.20.255 (144 addresses) is free for future labs. Lab A supports 62 hosts, Lab B 30 hosts and Lab C 14 hosts.

2. System 1 has IP 192.168.1.130 and System 2 has IP 192.168.1.190, both with subnet mask 255.255.255.192. Determine the subnet range, network address and broadcast address of each system, the valid host range, whether each IP falls in its subnet, and the usable hosts. (8 Marks)

Problem Understanding

Two hosts with mask 255.255.255.192 (/26) must be placed in their subnets, and their validity checked.

Method Selection

- Block size = 256 - 192 = 64, so subnets start at 0, 64, 128, 192 in the last octet.
- Network = largest multiple of 64 that is less than or equal to the host octet; broadcast = next network - 1.

Solution Process

- System 1: 130 lies between 128 and 191, so network = 192.168.1.128, broadcast = 192.168.1.191.
- System 2: 190 lies between 128 and 191, so it is in the same subnet, network = 192.168.1.128, broadcast = 192.168.1.191.
- Host bits = 6, usable hosts = 2^6 - 2 = 62.

System	IP address	Network address	Valid host range	Broadcast	In range?	Usable hosts
System 1	192.168.1.130	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191	Yes	62
System 2	192.168.1.190	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191	Yes	62

Interpretation of Results

Both IP addresses lie inside the range 192.168.1.128 - 192.168.1.191, so they belong to the **same subnet** and can communicate directly without a router. 192.168.1.190 is the last usable host address (192.168.1.191 is the broadcast), so it is valid. Each subnet supports 62 hosts.

3. An organization has the block 172.16.0.0/16. Subnet masks: HR /25, Sales /26, IT /27, Admin /28. Assign subnet addresses, find network address, host range, broadcast address, usable hosts, and the remaining unused range. (8 Marks)

Problem Understanding

A /16 block (65,536 addresses) is divided into four departmental subnets of different sizes (VLSM). Each subnet's details and the leftover space are required.

Method Selection

- Block sizes: /25 = 128, /26 = 64, /27 = 32, /28 = 16. Usable hosts = block size - 2.
- Allocate from the largest to the smallest, each starting where the previous one ends, to avoid overlap and wastage.

Solution Process

- HR: starts 172.16.0.0, ends at 172.16.0.127 (128 addresses).
- Sales: starts 172.16.0.128, ends at 172.16.0.191 (64 addresses).
- IT: starts 172.16.0.192, ends at 172.16.0.223 (32 addresses).
- Admin: starts 172.16.0.224, ends at 172.16.0.239 (16 addresses).

Dept	Subnet	Network address	Valid host range	Broadcast	Usable hosts
HR	/25	172.16.0.0	172.16.0.1 - 172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129 - 172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193 - 172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225 - 172.16.0.238	172.16.0.239	14

Interpretation of Results

Total addresses used = 128 + 64 + 32 + 16 = 240 (172.16.0.0 to 172.16.0.239). **Unused range: 172.16.0.240 to 172.16.255.255** (65,536 - 240 = 65,296 addresses), available for future expansion. Usable hosts: HR 126, Sales 62, IT 30, Admin 14.

4. A company uses 172.16.10.0/24 with a /27 subnet mask. Calculate the number of subnets and usable hosts per subnet, find the subnet of 172.16.10.78 with its network and broadcast address, and check whether 172.16.10.95 is in the same subnet. (8 Marks)

Problem Understanding

A /24 network is subnetted with /27 (mask 255.255.255.224). We must find subnet count, hosts per subnet, and locate two given addresses.

Method Selection

- Number of subnets = 2^n, where n = borrowed bits = 27 - 24 = 3.
- Hosts per subnet = 2^h - 2, where h = 32 - 27 = 5. Block size = 256 - 224 = 32.

Solution Process

- Subnets = 2^3 = 8. Usable hosts = 2^5 - 2 = 30.
- Subnet network addresses: .0, .32, .64, .96, .128, .160, .192, .224.
- 78 lies between 64 and 95, so 172.16.10.78 is in the subnet **172.16.10.64/27**.
- Network address = 172.16.10.64; broadcast = 172.16.10.95; host range = 172.16.10.65 - 172.16.10.94.

Item	Result
Total subnets	8
Usable hosts per subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Network address	172.16.10.64
Broadcast address	172.16.10.95
Valid host range	172.16.10.65 - 172.16.10.94

Interpretation of Results

172.16.10.78 is a valid host in the 172.16.10.64/27 subnet. 172.16.10.95 does fall within the range of that subnet (64 to 95), but it is the **broadcast address**, so it cannot be assigned to a host. The next subnet begins at 172.16.10.96.

5. A network uses the IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Compress it to its shortest form, expand it back, find the network prefix for /64, and calculate the total number of host addresses. (8 Marks)

Problem Understanding

An IPv6 address has 128 bits written as 8 groups of 16 bits (4 hex digits each). Compression, expansion, prefix extraction and host count are required.

Method Selection

- Rule 1: drop leading zeros in each group. Rule 2: replace the longest run of consecutive all-zero groups by '::' (only once).
- Network prefix = first 64 bits (first 4 groups). Host addresses = $2^{(128 - \text{prefix length})}$.

Solution Process

- Full address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Remove leading zeros: 2001:db8:0:0:0:ff00:42:8329
- Replace the three consecutive zero groups by '::' - **Compressed: 2001:db8::ff00:42:8329**
- Expand: restore leading zeros and the 3 missing zero groups (8 - 5 = 3) - **Expanded: 2001:0db8:0000:0000:0000:ff00:0042:8329**
- Prefix /64 = first four groups: 2001:0db8:0000:0000, so **network prefix = 2001:db8::/64**.
- Host bits = 128 - 64 = 64, so total addresses = 2^{64} = 18,446,744,073,709,551,616.

Item	Result
Compressed form	2001:db8::ff00:42:8329
Expanded form	2001:0db8:0000:0000:0000:ff00:0042:8329
Network prefix (/64)	2001:db8::/64
Interface ID (host part)	::ff00:42:8329
Total host addresses	2^{64} = 18,446,744,073,709,551,616

Interpretation of Results

The address belongs to the 2001:db8::/64 network, and the last 64 bits (0000:0000:ff00:0042:8329 portion, i.e. ::ff00:42:8329) identify the host. A /64 is the standard IPv6 LAN subnet size, providing about 1.8×10^{19} addresses, so address shortage is not a concern and no subnet-wide '-2' deduction is normally applied in IPv6.